ELSEVIER

FEMS Microbiology Letters 185 (2000) 163–168

FEMS MICROBIOLOGY Letters

www.fems-microbiology.org

The polyester polyurethanase gene (*pueA*) from *Pseudomonas chlororaphis* encodes a lipase

Robert V. Stern, Gary T. Howard *

Department of Biological Sciences, Southeastern Louisiana University, SLU 10736, Hammond, LA 70402, USA

Received 14 July 1999; received in revised form 15 December 1999; accepted 17 February 2000

Abstract

A gene (*pueA*, polyurethane esterase A) encoding an extracellular polyurethanase (PueA) was cloned from *Pseudomonas chlororaphis* into *Escherichia coli*. The enzyme secreted from *E. coli* showed esterase activity when assayed with *p*-nitrophenyl acetate. Subcloning of a 3.2-kb *Sal*I–*Eco*RI fragment into a T7 RNA polymerase expression vector (pT7-6) produced a ^{35}S-labeled protein of 65 kDa. Nucleotide sequencing of *pueA* showed an open reading frame encoding a 65-kDa protein of 617 amino acid residues, with the serine hydrolase consensus sequence GXSXG. PueA was over-expressed using the pT7-6 vector transformed into *E. coli* BL21(DE3) and was purified in one step using Sephadex G-75.

Keywords: Polyurethane; Polyurethanase; Sequence; Lipase; *Pseudomonas*

1. Introduction

As polyester-based polyurethane (PU) waste accumulates, methods are being sought to degrade and recycle this man-made, recalcitrant polymer. Biodegradation is an attractive avenue of investigation, since it has been discovered that microbial sources (bacteria [1–3] and fungi [4,5]) can utilize PU for growth. *Pseudomonas chlororaphis* can grow using PU as its sole carbon and energy source [2]. PU is used in the manufacturing of adhesives, thermal insulation, foam padding and paint [6], and is best described as a repeating carbamate polymer, -(R1-NH-CO_2-R2)-, which is susceptible to hydrolysis by esterase or protease attack.

Extracellular esterase activity is the primary activity found in both bacteria and fungi, which allows them to grow on PU. Six bacterial proteins [1,2,7–9] and one fungal protein [4] have been purified and five of these enzymes, polyurethanases, demonstrate esterase activity with little or no proteolytic activity. Recently, two of the bacterial polyurethanases have been cloned and expressed in *Escherichia coli* [10,11]. One of these enzymes is secreted from *Pseudomonas fluorescens* [10], while the other is found in *Comamonas acidovorans* [11] and is membrane-associated. The amino acid sequence derived from the nucleotide sequence of the *Comamonas* polyurethanase has revealed the serine hydrolyase motif, G-X-S-X-G, and showed a maximum amino acid sequence identity of 34% with carboxylesterases and acetylcholinesterases in GenBank. In this study, a gene (*pueA*) encoding a secreted polyurethanase from *P. chlororaphis* has been cloned, sequenced and its encoded protein over-expressed.

2. Materials and methods

2.1. Chemicals and bacterial strains

Amplify™ fluorographic reagent, [^{35}S]methionine (18.5 MBq, 500 μCi per 50 μl) and an ECL nucleic acid labeling kit were obtained from Amersham (Arlington Heights, IL, USA). λZAPII and Gigapack III Gold packaging extract were obtained from Stratagene (La Jolla, CA, USA). Enzyme substrates used were *p*-nitrophenyl acetate (ICN, Costa Mesa, CA, USA) and hide powder azure (Calbiochem, La Jolla, CA, USA). The nylon membrane for blotting was S&S Nytran from Schleicher and Schuell (Keene, NH, USA). Restriction enzymes and T4 DNA ligase were obtained from Sigma (St. Louis, MO, USA) or Gibco BRL (Grand Island, NY, USA). RPMI medium 1640

* Corresponding author. Tel.: +1 (504) 549-3501; Fax: +1 (504) 549-3851; E-mail: ghoward@selu.edu

0378-1097/00/$20.00
PII: S0378-1097(00)00094-X

(with glutamine, without methionine) and 1-kb DNA ladder were obtained from Gibco BRL (Grand Island, NY, USA). All other buffers and salts were reagent grade or better. *P. chlororaphis* was isolated from a microbial consortium obtained from the Naval Research Laboratory, Washington, DC, USA, by using its ability to degrade a water dispersible PU, Impranil DLN. *P. chlororaphis* was grown in Luria–Bertani (LB) medium at 30°C with constant shaking at 180 rpm. *E. coli* XL1-Blue MRF′ cells were obtained as part of the λZAPII kit and were grown in LB medium at 37°C with shaking at 220 rpm. *E. coli* BL21(DE3) and plasmids pT7-5 and pT7-6 were donated by Dr. Jessup Shively of Clemsen University.

2.2. Construction and screening of the genomic library

Chromosomal DNA was isolated by the method of Marmur [12]. The DNA was extracted two times with ether to remove trace organics [13]. λZAPII phage library was constructed as previously described [10]. Recombinant phages were screened by plating in soft agar overlays containing Impranil as previously described [10]. Plasmids containing *pueA* were excised and rescued from the λ library using the Stratagene protocol. The *E. coli* XL1-Blue were plated on LB agar plates supplemented with 1.5% (v/v) Impranil DLN dispersion and 60 μg ml^{-1} ampicillin. Plasmid DNA from the positive clones was isolated using alkaline lysis [13] and the restriction endonucleases *Eco*RI, *Kpn*I, *Sal*I, *Sst*I, *Cla*I, *Bgl*II, *Bam*HI, *Not*I, *Hind*II, *Xba*I, *Xho*I and *Eco*RV were tested. Results were detected using agarose gel electrophoresis with Tris–acetate buffer. Southern blot experiments were conducted as previously described [10].

2.3. Expression and purification of the polyurethanase

The 3.2-kb *Sal*I–*Eco*RI fragment of the insert from pPU2 was subcloned into the expression vectors pT7-5 and pT7-6, each with a T7 RNA promoter but at opposite ends of the inserted DNA. These plasmids were then used to transform competent *E. coli* C600 cells that contained the pGP1-2 plasmid encoding the T7 RNA polymerase. The procedure for the ^{35}S-labeling was previously described [10]. BL21(DE3) cells were made competent with dimethyl sulfoxide and transformed with the expression vector pT7-6 containing *pueA* as a 3.2-kb *Sal*I–*Eco*RI insert. These cells (25-ml culture) were grown at 37°C to an $OD_{600} = 0.6$ in LB medium containing 60 μg ml^{-1} ampicillin and 60 μg ml^{-1} chloramphenicol. At this time, isopropyl-β-D-thiogalactoside (IPTG, 200 mg ml^{-1} stock) was added to a final concentration of 1 mM. After 2 h, the cells were harvested by centrifugation and resuspended in 7 ml of 50 mM potassium phosphate buffer pH 7.0. The cells were lysed by adding 10% sodium dodecyl sulfate (SDS) to a final concentration of 1% SDS and warming the solution briefly at 50°C to solubilize the SDS. The

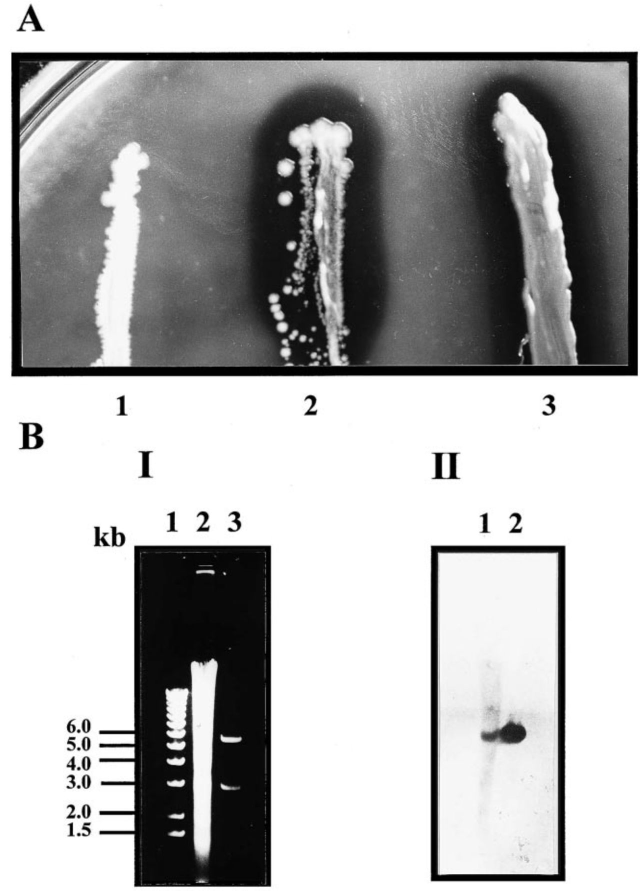

Fig. 1. (A) Polyurethanase activity detected as a zone of clearing on a LB/Impranil plate. (1) *E. coli* XL1-Blue transformed with pBluescript plasmid, (2) *E. coli* XL1-Blue transformed with pPU2 plasmid, (3) *P. chlororaphis*. The clearing observed in (2) and (3) indicates a polyurethanase enzyme is present. (B) Southern blot experiment demonstrating the 5.9-kb insert in the pPU2 plasmid was derived from *P. chlororaphis* chromosomal DNA. (I) Agarose gel electrophoresis pattern. Lane 1, 1-kb ladder; lane 2, *Eco*RI digest of *P. chlororaphis* chromosomal DNA; lane 3, *Eco*RI digest of pPU2 plasmid. (II) X-ray film exposure of blot from (I) probed with the 5.9-kb insert of pPU2 labeled with horseradish peroxidase.

lysate was treated with 0.1 mg DNase and allowed to stand overnight at 4°C to precipitate the SDS. One ml of the cell lysate was applied to a Sephadex G-75 column (45.0×1.0 cm) at a flow rate of 0.3 ml min^{-1} and 1-ml fractions were collected. One main peak was observed after the void volume as detected by SDS–polyacrylamide gel electrophoresis (SDS–PAGE) and this was the purified PueA. Polyurethanase activity was detected as a zone of clearing using a radial diffusion assay (RDA). The gel for the assay was made using 1.5% (w/v) agarose and 1.5% (v/v) Impranil DLN dispersion in 10 mM potassium phosphate pH 7.0. The wells held a maximum volume of 100 μl, and usually a zone of clearing would be evident in 6 h at 23°C which was dependent on enzyme concentration. For protease activity [10], insoluble hide powder azure (25 mg) was used as the substrate in 3 ml of 50 mM potassium phosphate pH 7.0, which was monitored at 595 nm. For esterase activity [10], 10 μl of *p*-nitrophenyl

GGG TTG CTG GGG GCC GGG ACA AGA TCA CGG GGT CGG CTT GAC CCT GGA CTG GCA GTT CTG AAG GCC GGG TAG CCA GCA GTC 81

CCC CGG CCC TTC ACC GAG TTG CCC GAG TCC CGC CAT GGG GCT TGG GCG GCA TGC GTT TGC TCT GAC AAT TCC AAC AAC AG 161

AAGAGGC AAT AAC ATG GGT GTG TTT GAC TAC AAG AAC TTC ACT GCA AGT GAC TCC AAG GCG CTG TTT TCC GAT GCG CTG GCG 243
S.D. M G V F D Y K N F T A S D S K A L F S D A L A

ATC ACC CTG TAC TCC TAC CAC AAC ATC GAC AAC GGC TTT GCC GAG GGC TAC CAG CAC AAC GGC TTC GGC CTG GGC CTG CCG 324
I T L Y S Y H N I D N G F A E G Y Q H N G F G L G L P 50

GCG ACC CTG GTC ACG GCG CTG ATC GGC AGC GGC AAT TCC CAG GGG GTA ATC CCC GGC ATT CCC TGG AAC CCC GAC TCC GAG 405
A T L V T A L I G S G N S Q G V I P G I P W N P D S E

AAG GCC GCG CTG GAT GCC CTG CAC CAG GCC GGC TGG TCC ACC ATC AGC GCC CAG CAA CTG GGC TAC GAC GGC AAG GTC GAT 486
K A A L D A L H Q A G W S T I S A Q Q L G Y D G K V D 104

GGC CGG GGA ACC TTC TTT GGC GAG AAG GCC GGC TAC GGC ACG GCC CAG GTG GAA ATC CTC GGC AAG TAC GAT GCT CAG GGC 567
G R G T F F G E K A G Y G T A Q V E I L G K Y D A Q G

CAC CTG GAA TCC ATC GGC ATT GCC TTT CGC GGC ACC AGC GGG CCA CGG GAA TCG GTG ATC AGT GAC ACC ATC GGT GAT GTG 648
H L E S I G I A F R G T S G P R E S V I S D T I G D V 158

ATC AAC GAC CTG CTG GCG GCC CTG GGG CCC AAG GAC TAT GCG AAG AAC TAT GCC GGC GAG GCG TTC GGC AAG CTG CTG GGA 729
I N D L L A A L G P K D Y A K N Y A G E A F G K L L G

GAT GTC GCC GCC TTC GCC CAG GCC AAT GGC CTG TCG GGC AAG GAT GTC CTG GTC AGC GGC CAC AGC CTG GGC GGG CTG GGG 810
D V A A F A Q A N G L S G K D V L V S G H S L G G L G 212

GTC AAC AGC CTG GCG GAC CTG AGC AGC GAA CGC TGG TCG GGA TTC TAC AAG GAC TCC AAC TAC ATC GCC TAC GCC TCG CCG 891
V N S L A D L S S E R W S G F Y K D S N Y I A Y A S P

ACC CAG AGC GCC AGC GAC AAG GTG CTG AAC ATC GGC TAC GAG AAC GAC CCG GTG TTC CGC GCC CTG GAC GGC TCA ACC TTC 972
T Q S A S D K V L N I G Y E N D P V F R A L D G S T F 266

AAC CTG TCG TCC CTC GGG GTG CAT GAC GCC CAC CAG GAC TCG GCG ACC AAC AAC ATC GTC AAC TTC AAC GAC CAT TAC GCC 1053
N L S S L G V H D A H Q D S A T N N I V N F N D H Y A

TCG ACC CTG TGG AAT GTG CTG CCG TCC TCG ATC CTC AAC ATT CCC ACC TGG CTC TCC CAC CTG CCC ACC GGC TAC GGC GAC 1134
S T L W N V L P S S I L N I P T W L S H L P T G Y G D 320

GGC CTG TCC CGG GTG CTG GAG TCG AAG TTC TAC GAC TTC ACC AGC AAG GAC TCC ACC ATC GTG GTA GCC AAC CTG TCG GAC 1215
G L S R V L E S K F Y D F T S K D S T I V V A N L S D

CCG GCC CGG GCC AGT ACC TGG GTC CAG GAC CTC AAC CGC AAT GCC GAG ACC CAC AAG GGC AGC ACC TTC ATC ATC GGC AGC 1296
P A R A S T W V Q D L N R N A E T H K G S T F I I G S 374

GAC GGC AAC GAC CTG ATC CAG GGC GGC AGC GGC AAC GAC TAC CTG GAA GGC CGG GCC GGC AAC GAT ACG TTC CGC GAC AGC 1377
D G N D L I Q G G S G N D Y L E G R A G N D T F R D S

GGC GGC TAC AAC ATC ATC CTC GGC GGC CAG GGC AGC AAC ACC CTG GAC CTG CAG CAG TCG GTC AAG AAC TAC AGC TTT GCC 1458
G G Y N I I L G G Q G S N T L D L Q Q S V K N Y S F A 428

AGC GAC GGC GCC GGG ACC CTG TAC CTG CGT GAC GCC AAT GGC GGT ATC AGC ATG ACC CGC GAC ATT GGC GCG ATC CAG AGC 1539
S D G A G T L Y L R D A N G G I S M T R D I G A I Q S

AAG GAG CCG GGC TTT CTC TGG GGC CTG TTC AAG GAC GAT GTG ATC CAC CAG GTC ACC GAC CAG GGC CTC AAG GCC GGC GGG 1620
K E P G F L W G L F K D D V I H Q V T D Q G L K A G G 482

CAG CTT ACC CAG TAC GCC TCG TCG GTC AGG GGC GAT GCG GGC GAC AAC GTG CTC AAG GCC CAT GCC GGC GGC GAC TGG CTG 1701
Q L T Q Y A S S V R G D A G D N V L K A H A G G D W L

TTC GGC CTG GAC GGC AAC GAC CAC CTG ATC GGC GGC CAG GGC AAC GAC GTG TTC GTT GGC GGT GCC GGC AAC GAC CTG ATG 1782
F G L D G N D H L I G G Q G N D V F V G G A G N D L M 536

GAA GCC GGA GGC GGC AAC AAT ACC TTT CTG TTC ACC GGG CAT TTT GGC CAG GAC CGC ATC CTC GGC TAC CAG GGT GGC GAC 1863
E A G G G N N T F L F T G H F G Q D R I L G Y Q G G D

AAG CTG GTG TTC ATG GGG GTG AGC GGC GTG CTG CCG GAC CAG GAC TAT CGG GCT CAT GCC AGC AGC AGT GGT AAC GAC ACG 1944
K L V F M G V S G V L P D Q D Y R A H A S S S G N D T

GTG CTG ACC TTC GGC CAG GAC TCG GTG ACC CTG GTG GGG GTC AGC CTG GAG CAC CTC AAC GGT TCC GGC ATC GTT CTC GCC 2025
V L T F G Q D S V T L V G V S L E H L N G S G I V L A 617

TGA TGC CGG CGC GGC GCG AGG CCT GCA ATG GGC CTC GCA CCG TTC ATT GCC GGG GGA AAG TGA CCT GCC AGT CGA TTT 2103

Fig. 2. Nucleotide sequence of the *P. chlororaphis* polyurethanase gene (*pueA*) and the deduced amino acid sequence for the encoded enzyme. The putative active site containing the sequence GXSXG and the secretion signal GGXGXDXXX are underlined. The proposed Shine–Dalgarno site for ribosomal binding is indicated in bold (S.D.). The nucleotide sequence is available in GenBank, accession #AF069748.

acetate (60 μg per 100 ml acetonitrile) was used as the substrate in a reaction mixture of 2 ml 50 mM potassium phosphate pH 7.0 and 1 ml acetonitrile. The reaction was monitored at 420 nm. Protein concentrations were estimated by the method of Lowry [14] using the Bio-Rad DC Protein Assay. SDS–PAGE was performed as described by Laemmli [15] using a 10% gel. Prestained standards (Gibco BRL, Sigma) or Dalton Mark VII-L standards (Sigma) were used. Following electrophoresis, gels were stained with Coomassie blue R-250. For the

A

Organism		Active Site										Identity	Accession #
Pseudomonas fluorescens SIK W1	202-	Val	Ser	Gly	His	Ser	Leu	Gly	Gly	Leu	-210	78%	D11455
Pseudomonas fluorescens B52	203-	Val	Ser	Gly	His	Ser	Leu	Gly	Gly	Leu	-211	80%	M86350
Pseudomonas LS107d2	203-	Val	Ser	Gly	His	Ser	Leu	Gly	Gly	Leu	-211	73%	M74125
Serratia marcescens SM6	203-	Val	Ser	Gly	His	Ser	Leu	Gly	Gly	Leu	-211	64%	U11258
Serratia marcescens Sr41	202-	Ile	Ser	Gly	His	Ser	Leu	Gly	Gly	Leu	-210	64%	D13253
Pseudomonas chlororaphis	203-	Val	Ser	Gly	His	Ser	Leu	Gly	Gly	Leu	-211	100%	AF069748
Consensus sequence				Gly	X	Ser	X	Gly					

B

Organism		Secretion Signal	
Pseudomonas fluorescens SIK W1	363-	E P H T G N T F I I G S D G N D L I Q G G K G A D F I E G G K G N D T I R	-399
Pseudomonas fluorescens B52	363-	E P H K G N T F I I G S D G N D L I Q G G N G A D F I E G G K G N D T I R	-399
Pseudomonas LS107d2	363-	E P H K G D T F I I G S A G N D L I Q G G K G A D F I E A G K G N D T I R	-399
Serratia marcescens SM6	364-	E T H S G P T F I I G S D G N D L I K G G K G N D Y L E G R D G D D I F R	-400
Serratia marcescens Sr41	363-	E T H S G P T F I I G S D G N D L I K G G K G N D Y L E G R D G D D I F R	-399
Pseudomonas chlororaphis	363-	E T H K G S T F I I G S D G N D L I Q G G S G N D Y L E G R A G N D T F R	-399
Consensus sequence		G G X G X D X X X	

Fig. 3. Comparison of putative active sites and secretion signal sites for the *P. chlororaphis* polyurethanase and five extracellular lipases found in GenBank. The numbers indicate the position of the amino acid residues within the protein sequence. (A) Active site region and the overall amino acid identity of the ORF indicated as a percentage. (B) Secretion signal region with residues identical to the *P. chlororaphis* sequence shaded.

^{35}S-labeling experiment, the gel was fixed for 30 min, soaked in Amplify™ for 30 min, dried and autoradiographed with Kodak X-OMAT XAR-5, X-ray film for 45 h at −80°C.

2.4. Nucleotide analysis and N-terminal analysis

The nucleotide sequence and N-terminal protein sequence were determined by automated microsequencing performed in the Sequencing Laboratory of the Biotechnology Center at the University of Illinois at Urbana-Champaign. Both DNA strands were sequenced and the alignment was deduced using Sequencher 3.1™ (Gene Codes Corporation, Ann Arbor, MI, USA). The open reading frame (ORF) and the amino acid sequence were also determined using Sequencher 3.1™. Protein blotting for N-terminal analysis was performed using 25 mM Tris, 192 mM glycine, 20% methanol (v/v), pH 8.3, transfer buffer and a mini transblot electrophoretic transfer cell (Bio-Rad, Hercules, CA, USA). Sequi-Blot PVDF membrane was used (Bio-Rad).

3. Results

3.1. Screening the λ library for pueA

The genomic library prepared by ligating 4–10-kb *Eco*RI fragments of *P. chlororaphis* chromosomal DNA into the *Eco*RI site of λZAPII was screened for polyurethanase activity based on the ability of the enzyme to hydrolyze PU. The hydrolysis was evident as a readily distinguishable plaque which was more translucent than neighboring plaques since the milky-white PU, Impranil, was hydrolyzed. A total of 4000 recombinants were screened, and 20 polyurethanase-positive clones were isolated. Although plaques were evident after an overnight incubation at 37°C, the polyurethanase activity did not appear until the plates were transferred to 23°C for several hours.

3.2. Cloning pueA in E. coli

Using the bacteriophage from several representative clones, plasmid excision and rescue were carried out and the resulting pBluescript plasmids were harbored in *E. coli* XL1-Blue. *E. coli* XL1-Blue containing pPU2 (5.9-kb insert of chromosomal DNA in pBluescript) were plated along with *P. chlororaphis* and a control of *E. coli* XL1-Blue cells containing pBluescript on an LB/Impranil plate (Fig. 1A). There were strong zones of clearing for the *E. coli* containing pPU2 and *P. chlororaphis*, while the *E. coli* containing pBluescript showed no clearing. Using pPU2, the restriction endonucleases *Kpn*I, *Sal*I and *Sst*I were found to cut the 5.9-kb insert within a narrow region that divided the insert approximately in half. *Cla*I, *Bgl*II, *Bam*HI, *Not*I, *Hin*dIII, *Xba*I, *Xho*I and *Eco*RV did not cut the 5.9-kb insert. To insure that the insert in pPU2 was from the *P. chlororaphis* chromosome, Southern blot hybridization was conducted (Fig. 1B). A chemi-luminescent probe was constructed from the 5.9-kb insert by labeling it with horseradish peroxidase. The probe hybridized strongly to the 5.9-kb insert and to the same-sized fragments from the chromosomal DNA indicating that the insert had originated from *P. chlororaphis* and that only one copy of the gene was present on the chromosome.

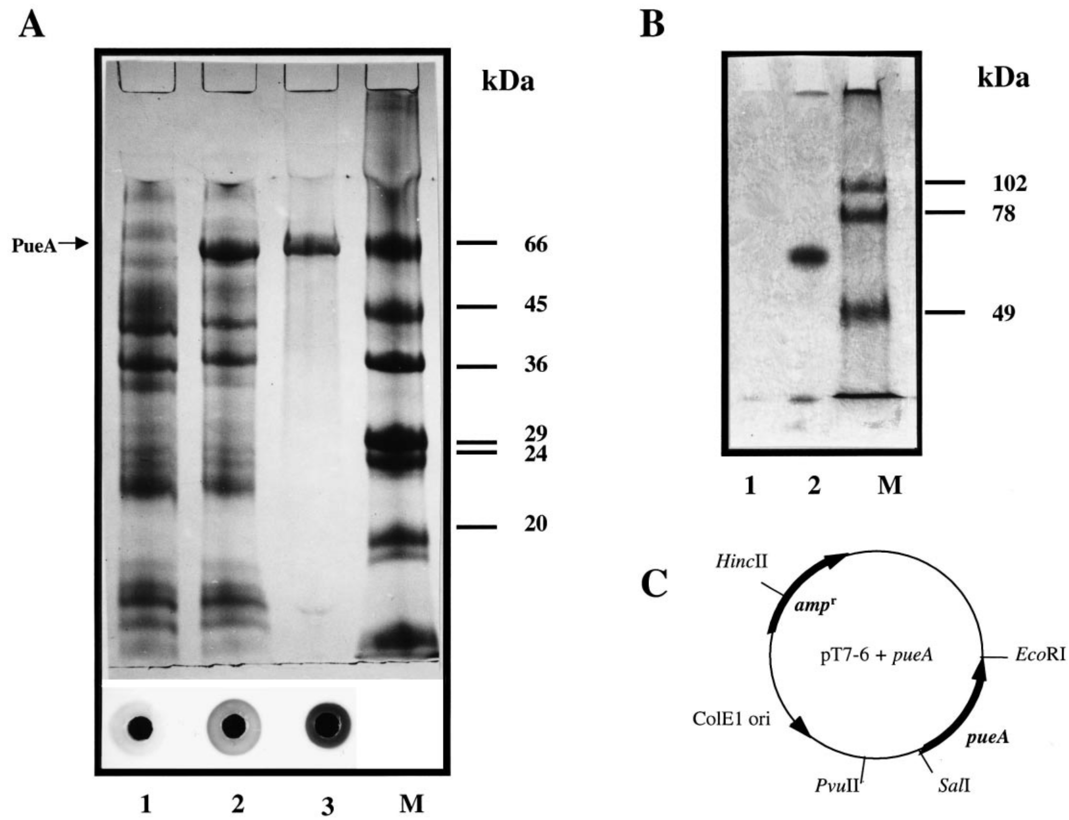

Fig. 4. Expression of the 65-kDa polyurethanase and the expression vector (pT7-6) used. (A) Over-expression of the 65-kDa polyurethanase from *P. chlororaphis* demonstrated using SDS–PAGE and a RDA for each lane. Lysed *E. coli* BL21(DE3) cells containing the expression vector (pT7-6) with *pueA* as the insert were compared before and after induction. Lane 1, control, no IPTG, 27 μg protein; lane 2, over-expression, 1 mM IPTG for 2 h, 24 μg protein; lane 3, PueA after passage over Sephadex G-75, 12 μg protein; lane M, Dalton Mark VII-L molecular mass markers. The RDA below each lane indicates the polyurethanase activity as a zone of clearing. Lanes 1 and 2 (0.4 mg protein each), lane 3 (0.08 mg protein). (B) Autoradiograph of the protein expressed in the [^{35}S]methionine labeling experiment using the *Sal*I–*Eco*RI fragment subcloned into pT7-5 and pT7-6. Lane 1, pT7-5 result (10 μl loaded); lane 2, pT7-6 result (10 μl loaded); lane M, molecular mass markers. (C) Expression vector pT7-6, 2210 bp, with *pueA* insert, 3200 bp. The T7 promoter is present between the *Pvu*II and *Sal*I sites.

3.3. Nucleotide sequence of pueA and the translated ORF

After sequencing *pueA*, an ORF of 1851 nucleotides encoding a protein, PueA, of 617 amino acids with a molecular mass of 65 kDa was found (Fig. 2). In order to confirm that the correct ORF had been found, the protein was blotted on PVDF membrane and its N-terminus was determined to be MGVFDYKNFTAS, which is an exact match for the putative ORF. Further analysis of the ORF showed two regions of special interest, one a putative active site and the other a region found to be conserved in secreted proteins that lack an N-terminal leader sequence (Fig. 2, underlined). The serine hydrolase motif, G-X-S-X-G, was found at Gly$_{205}$, while the secretion consensus signal was found at Gly$_{382}$. The active site and secretion signal sequences align well with conserved regions within extracellular lipases found by using a BLASTP search of GenBank at the National Center for Biotechnology Information (Fig. 3). Not only are the reported sequences highly conserved, but they are also all located within the same region of the lipase. The active site residues are found from residue 202 to 211 and the secretion signal residues are found from residue 363 to 400 for all the lipases (Fig. 3).

3.4. Subcloning of pueA and expression

In order to narrow the search for the polyurethanase gene, the 5.9-kb insert was digested with *Sal*I which yielded 2.7-kb and 3.2-kb fragments which were subcloned into the expression vectors pT7-5 and pT7-6. A protein of approximately 65 kDa was expressed by the pT7-6 plasmid as observed by using the radiolabel, [^{35}S]methionine (Fig. 4B, lane 2). No expression was observed for the pT7-5 plasmid (Fig. 4B, lane 1) which indicates that the gene is read from *Sal*I to *Eco*RI. In order to attempt to classify the extracellular protein PueA as an esterase or a protease, two substrates were tested. No activity was observed for extracellular protein using hide powder azure, which is a known substrate for the protease trypsin. Esterase activity was observed using *p*-nitrophenyl acetate for the case of extracellular protein from *E. coli* harboring pPU2 which was 1.5 times greater than the activity for the control cells. Over-expression of the enzyme, by transforming *E. coli*

BL21(DE3) with the pT7-6 expression vector containing the 3.2-kb *Sal*I–*Eco*RI insert, facilitated obtaining enough protein for conducting N-terminal sequencing. Induction of the protein was achieved with 1 mM IPTG for 2 h. In one chromatographic step (Sephadex G-75), good yields of the purified enzyme were obtained (Fig. 4A, lane 3). To confirm expression of active PueA, a RDA was conducted (Fig. 4A). A RDA using spirit blue agar (Difco) and Wesson oil confirmed that lipase activity was also present (not shown).

4. Discussion

Evidence has been presented which indicates the polyurethanase, PueA, from *P. chlororaphis* is a lipase. Using the deduced amino acid sequence, a BLASTP search of GenBank found five lipases with 60–80% identity to PueA. The active site and secretion signal regions are also highly conserved in these lipases. To further corroborate the conclusion PueA is a lipase, the polyurethanase enzyme from *P. fluorescens* which had been cloned [10] has now been sequenced (accession #AF144089, enzyme designated PulA for PU lipase A) and it showed amino acid sequence identity of 50–70% to lipases in GenBank. Comparing the amino acid sequence of PulA to PueA yielded 56% identity.

The ORF of *pueA* was confirmed by comparison of the deduced N-terminal amino acid sequence and the experimentally determined N-terminus, which yielded an exact match. A search upstream at nucleotides 162–167 revealed a sequence AAGAGG, which is the probable ribosomal binding site. The deduced amino acid sequence yields a molecular mass for PueA of 65 kDa that corresponds exactly with the [^{35}S]methionine-labeled protein observed and the protein observed after IPTG-induced over-expression (Fig. 4). A 63-kDa, native polyurethanase had been purified from *P. chlororaphis* [8] and PueA probably represents the same enzyme. Of the remaining 19 polyurethanase-active clones, only one was found which had a different restriction enzyme map than *pueA* and it is being studied.

While the five related lipases varied in total amino acid residues, *P. fluorescens* SIK W1 (449), *Pseudomonas* LS107d2 (474), *P. fluorescens* B52 (476) and the *Serratia* strains (both 613), the active site and the secretion signal sequences aligned perfectly, indicating this is a well conserved enzyme. One interesting note that may further distinguish PueA is that it lacks the amino acid cysteine, and four of the five related lipases also contain no cysteine.

Another polyurethanase [11] whose sequence is in GenBank (accession #AB00906) is from *C. acidovorans* and is membrane-associated. This enzyme is more closely related to carboxylesterases and acetylcholinesterases than to lipases. In conclusion, there appear to be a least two mechanisms for serine hydrolase activity that can degrade polyester-based PU.

Acknowledgements

This work was supported by the Office of Naval Research Grant N00014-96-1229.

References

[1] Howard, G.T. and Blake, R.C. (1998) Growth of *Pseudomonas fluorescens* on a polyester-polyurethane and the purification and characterization of a polyurethanase-protease enzyme. Int. Biodeterior. Biodegrad. 42, 213–220.

[2] Howard, G.T., Ruiz, C. and Hilliard, N.P. (1999) Growth of *Pseudomonas chlororaphis* on a polyester-polyurethane and the purification and characterization of a polyurethanase-esterase enzyme. Int. Biodeterior. Biodegrad. 43, 7–12.

[3] Nakajima-Kambe, T., Onuma, F., Kimpara, N. and Nakahara, T. (1995) Isolation and characterization of a bacterium which utilizes polyester polyurethane as a sole carbon and nitrogen source. FEMS Microbiol. Lett. 129, 39–42.

[4] Crabbe, J.R., Campbell, J.R., Thompson, L., Walz, S.L. and Schwultz, W.W. (1994) Biodegradation of a colloidal ester-based polyurethane by soil fungi. Int. Biodeterior. Biodegrad. 33, 103–113.

[5] Pathirana, R.A. and Seal, K.J. (1984) Studies on polyurethane deteriorating fungi. Part 1. Isolation and characterization of the test fungi employed. Int. Bioderterior. 20, 163–168.

[6] Battye, W. (1997) Best management practices for pollution prevention in the slabstock and molded flexible polyurethane foam industry: manual. U.S. Environmental Protection Agency, Office of Research and Development, National Risk Management Research Laboratory Cincinnati, EPA/625/R-96/005, Cincinnati, OH.

[7] Allen, A.B., Hilliard, N.P. and Howard, G.T. (1999) Purification and characterization of a soluble polyurethane degrading enzyme from *Comamonas acidovorans*. Int. Biodeterior. Biodegrad. 43, 37–41.

[8] Ruiz, C., Main, T., Hilliard, N.P. and Howard, G.T. (1999) Purification and characterization of two polyurethanase enzymes from *Pseudomonas chlororaphis*. Int. Biodeterior. Biodegrad. 43, 43–47.

[9] Akutsy, Y., Nakajima-Kambe, T., Nomura, N. and Nakahara, T. (1998) Purification and properties of a polyester polyurethane degrading enzyme from *Comamonas acidovorans* TB-35. Appl. Environ. Microbiol. 64, 62–67.

[10] Vega, R.E., Main, T. and Howard, G.T. (1999) Cloning and expression in *Escherichia coli* of a polyurethane-degrading enzyme from *Pseudomonas fluorescens*. Int. Biodeterior. Biodegrad. 43, 49–55.

[11] Nomura, N., Shigeno-Akutsu, Y., Nakajima-Kambe, T. and Nakahara, T. (1998) Cloning and sequence analysis of a polyurethane esterase of *Comamonas acidovorans* TB-35. J. Ferment. Bioeng. 86, 339–345.

[12] Marmur, J. (1961) A procedure for the isolation of deoxyribonucleic acid from micro-organisms. J. Mol. Biol. 3, 208–218.

[13] Sambrook, J., Fritsch, E. and Maniatis, T. (1989) Molecular Cloning: a Laboratory Manual, 2nd edn. Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY.

[14] Lowry, O.H., Rosebrough, N.J., Farr, A.L. and Randall, R.J. (1951) Protein measurement with the Folin phenol reagent. J. Biol. Chem. 193, 265–275.

[15] Laemmli, U.K. (1970) Cleavage of structural proteins during the assembly of the head of bacteriophage T4. Nature 227, 680–685.